

GAIRA V7 — Phase 04

Frozen Projection Engine and Hierarchical Inference

Figure report — all 14 figures with captions

Status	GATE_FAILED — 10 of 11 gates; OOD on real SERS fails
Engine version	v7_engine_v1
Frozen atlas / LSM / CSM / theme	all four verified before any computation
Projection	elastic_net
LSM to CSM	direct_csm_projection
Theme mode	confidence_weighted
BSV	theme_only, dimension 4
BSV effective rank	2.40 of 4
Geometry extension	landmark_barycentric
Held-out molecule top-1 (CSM)	0.799
Held-out class top-1, unseen molecule	0.855 (raw spectrum 0.608)
Dictionary leakage	+0.055 top-1
OOD AUROC (real Ag-SERS)	0.548 — FAILS
Confidence calibration	ECE 0.486 — poor
Completed	2026-08-06 16:37:17 UTC

Committed sources of record: reports/PHASE_04_REPORT.md and reports/PHASE_04_SCIENTIFIC_AUDIT.md
Projection only: nothing in Phases 00-03 was refitted, recomputed or replaced.
Figures are embedded from the committed PNGs, so this PDF cannot drift from the run that produced them.

Figures

01	inference pipeline	14	architecture
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Figure 01

The canonical GAIRA inference pathway



projection only — no fitting, no RNG, batch-independent, bit-identical on re-run

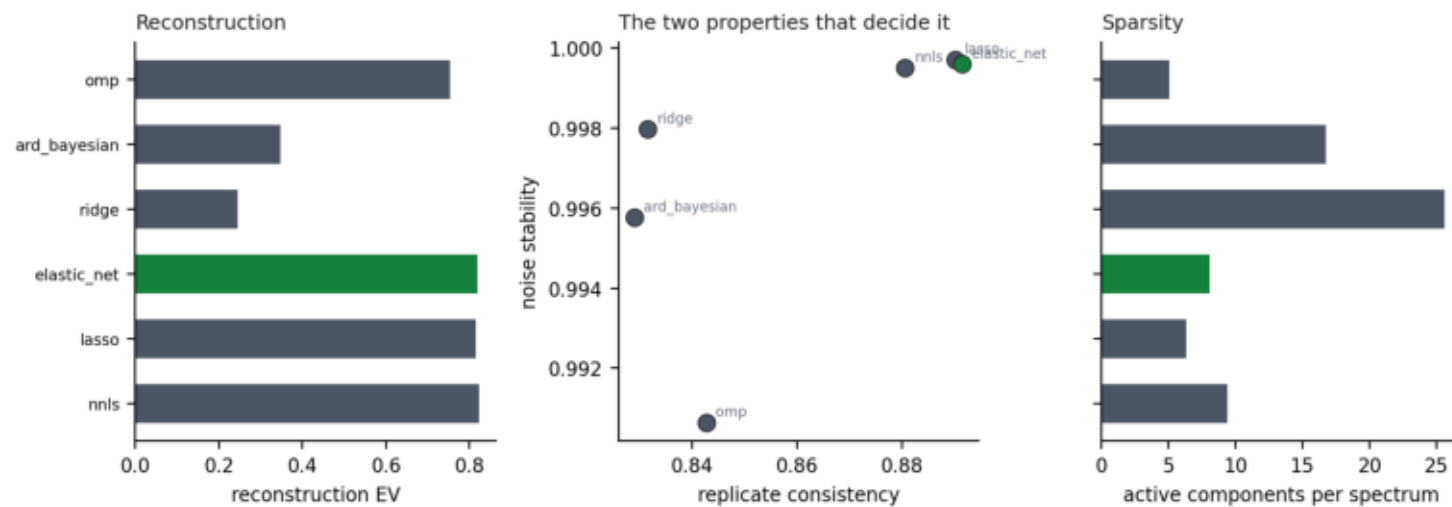
projection: elastic_net · aggregation: direct_csm_projection · theme mode: confidence_weighted · bsv variant: theme_only · geometry extension: landmark_barycentric

every output traces back: theme → CSM → LSM → canonical molecule → source spectrum

The canonical inference pathway. Projection only — no fitting, no random number, batch-independent, bit-identical on re-run. Every output traces back to named canonical molecules.

Figure 02

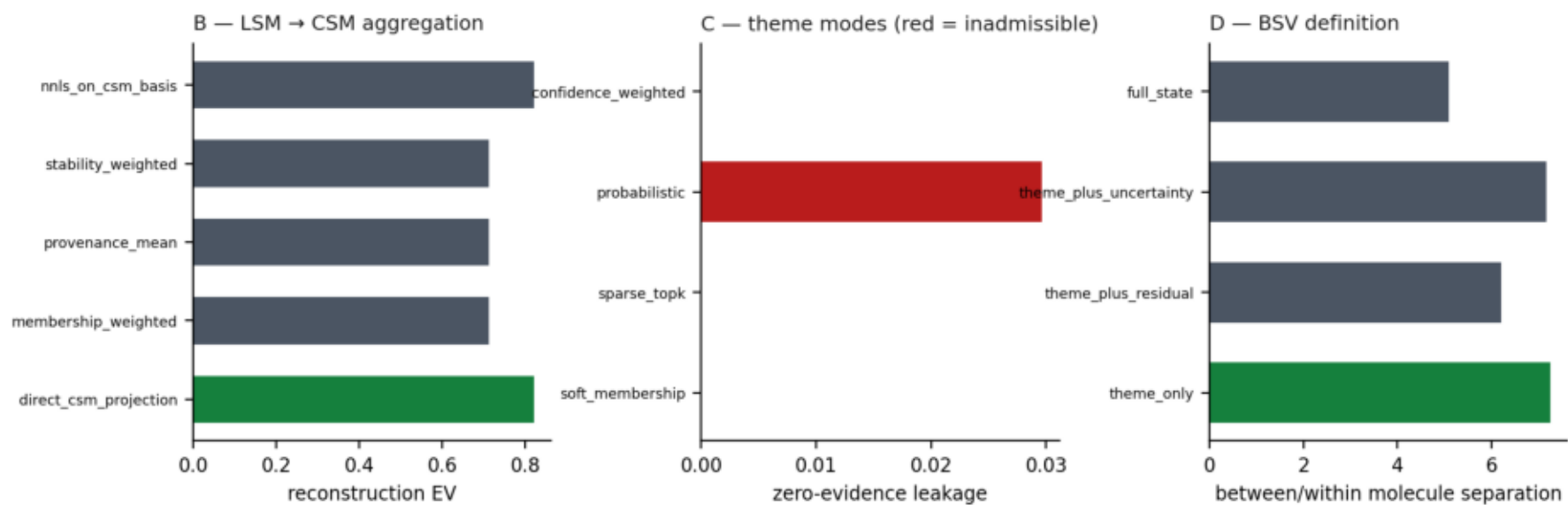
Projection estimators. Selected: elastic_net — chosen on replicate consistency x noise stability among estimators with zero negative mass, not on reconstruction alone.



Projection estimators. Elastic net selected on replicate consistency x noise stability among estimators with zero negative mass, not on reconstruction alone. Ridge and ARD reconstruct at 0.24 and 0.35 — an unconstrained fit on a coherent dictionary is not usable.

Figure 03

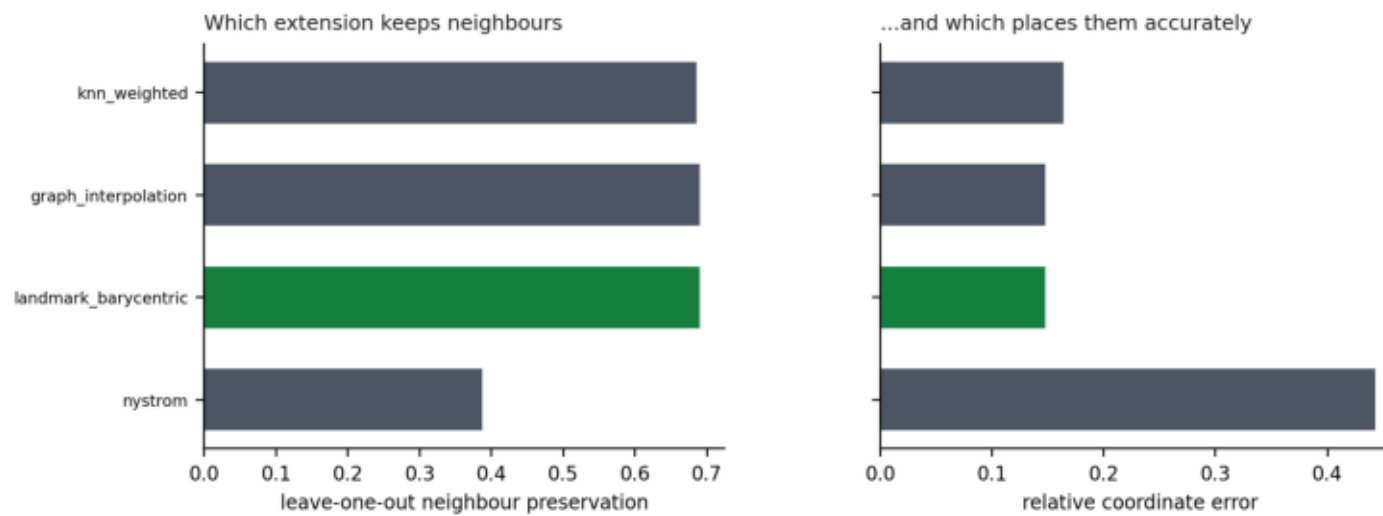
Aggregation, theme mode and BSV definition. The softmax mode scored best on replicate consistency and was rejected: it activates themes for which the spectrum has no CSM evidence at all.



Aggregation, theme mode and BSV definition. The softmax theme mode scored BEST on replicate consistency (0.993) and was rejected: it activates themes for which the spectrum has no CSM evidence at all. Adding uncertainty channels INTO the BSV lowers molecule separation, so they stay beside it.

Figure 04

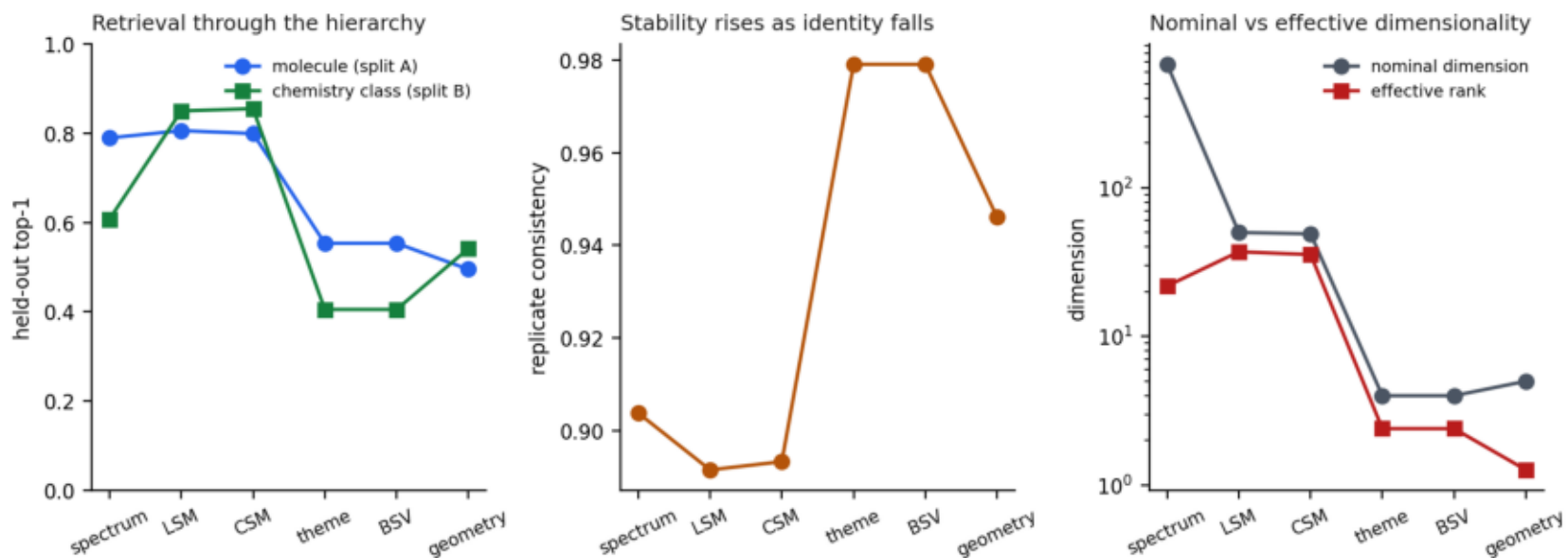
Out-of-sample manifold extension. Selected: landmark_barycentric. Nystrom is the principled choice for a diffusion map and the worst here — the frozen coordinates are too few and too spread for a kernel average to localise.



Out-of-sample manifold extension. Nystrom is the principled choice for a diffusion map and the worst here — 49 reference coordinates are too few and too spread for a kernel average to localise.

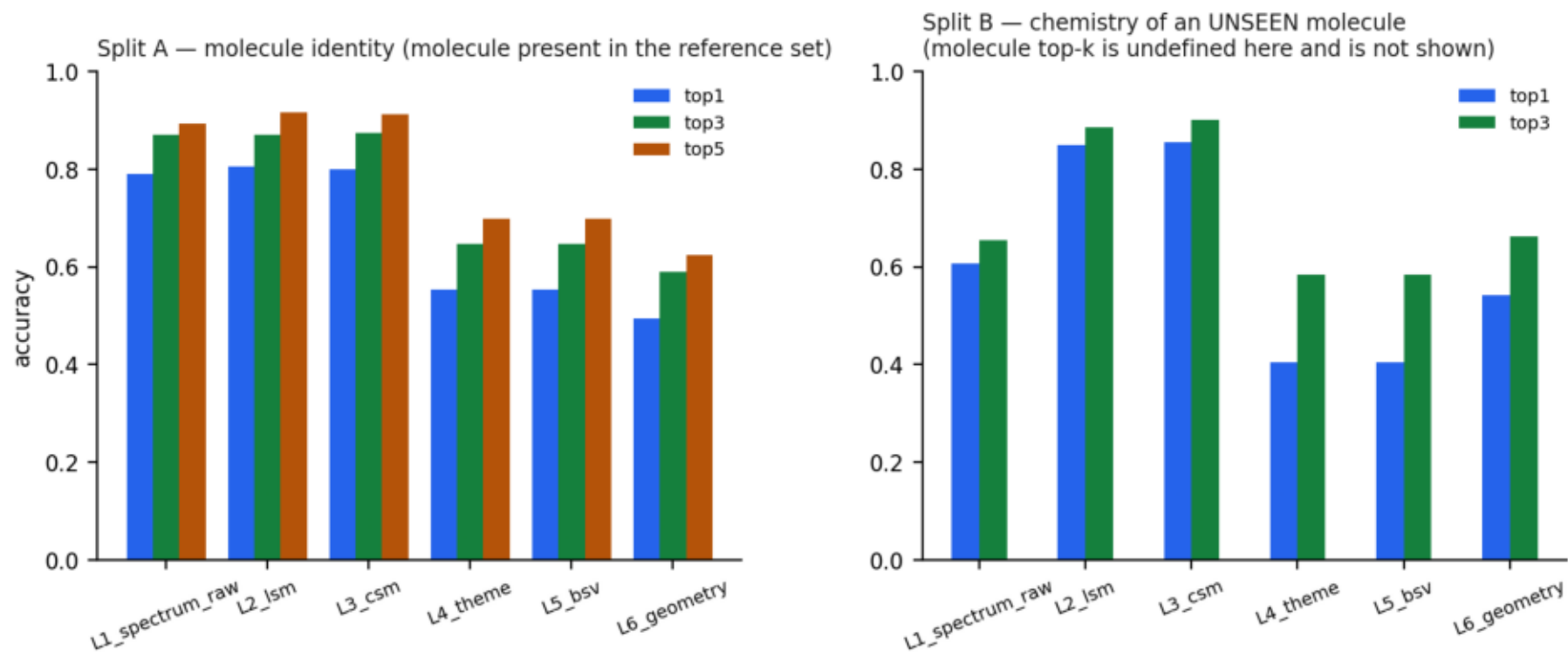
Figure 05

Information flow. The LSM/CSM layers RAISE chemistry generalisation to unseen molecules ($0.608 \rightarrow 0.855$); the theme layer trades both retrieval axes for the highest replicate consistency in the stack.



THE CENTRAL RESULT. The LSM/CSM layers raise chemistry-class retrieval on UNSEEN molecules from 0.608 to 0.855 — transfer, not recall. The theme layer trades both retrieval axes for the highest replicate consistency in the stack. Effective rank falls far below nominal dimension at every level.

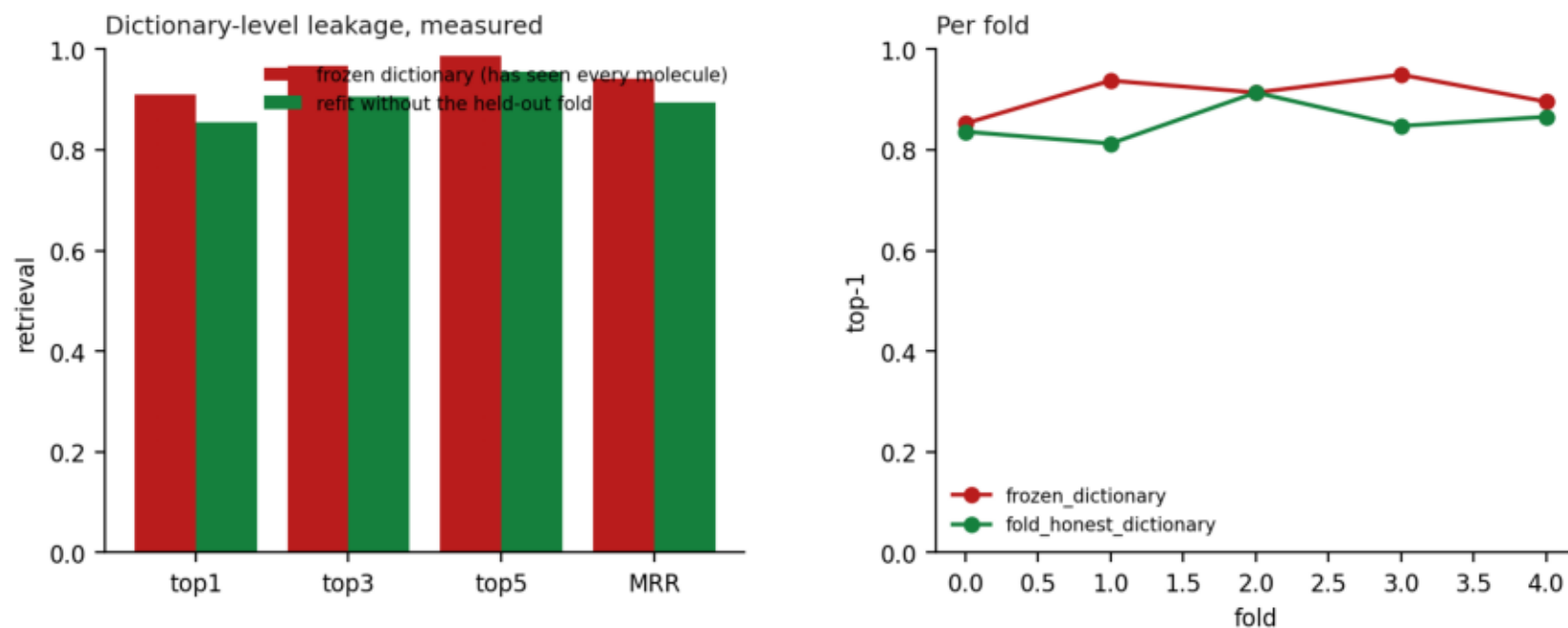
Figure 06



Retrieval by level under both splits. Split A holds out one spectrum with its molecule still represented; split B holds out the whole molecule, where molecule top-k is undefined by construction and is not shown.

Figure 07

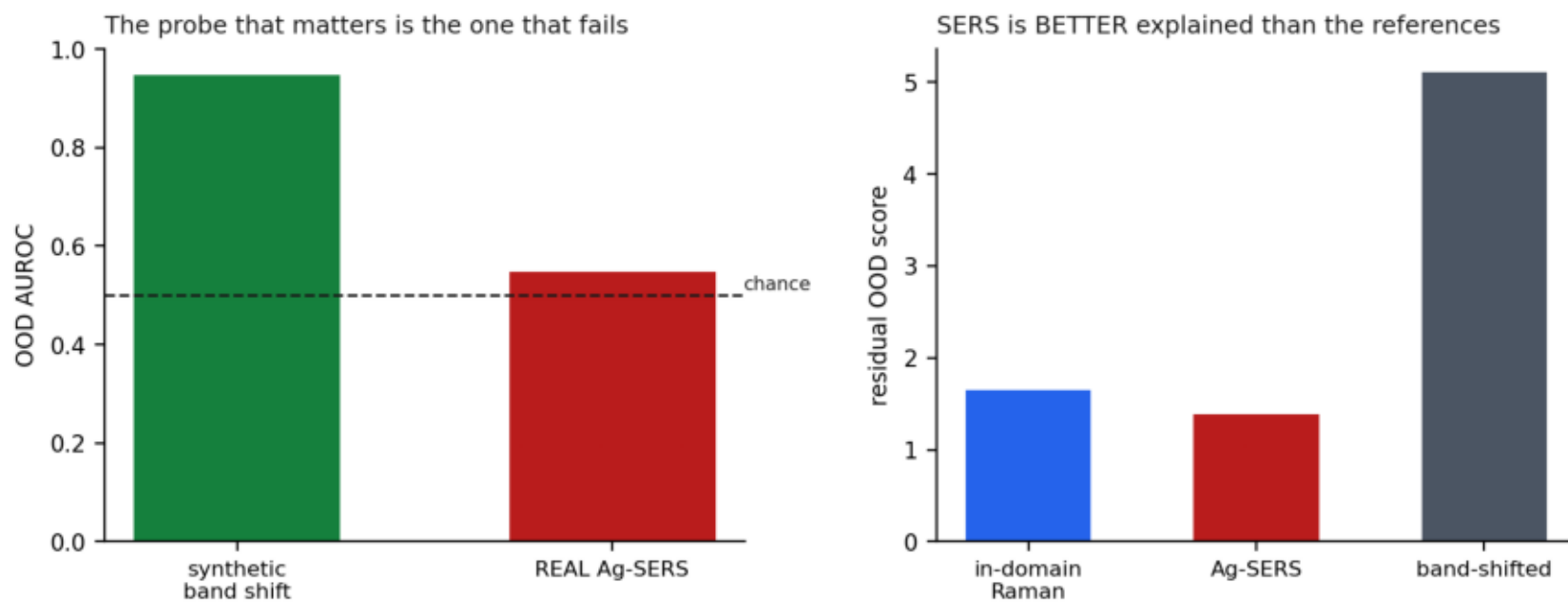
The frozen dictionary was fitted on every molecule, so grouping the folds cannot remove dictionary-level leakage. Refitting per fold measures it: +0.055 top-1 inflation in every in-sample number this project has produced.



Dictionary-level leakage, measured rather than assumed. The frozen dictionary was fitted on every molecule, so grouping the folds cannot remove it; refitting per fold quantifies it at +0.055 top-1.

Figure 08

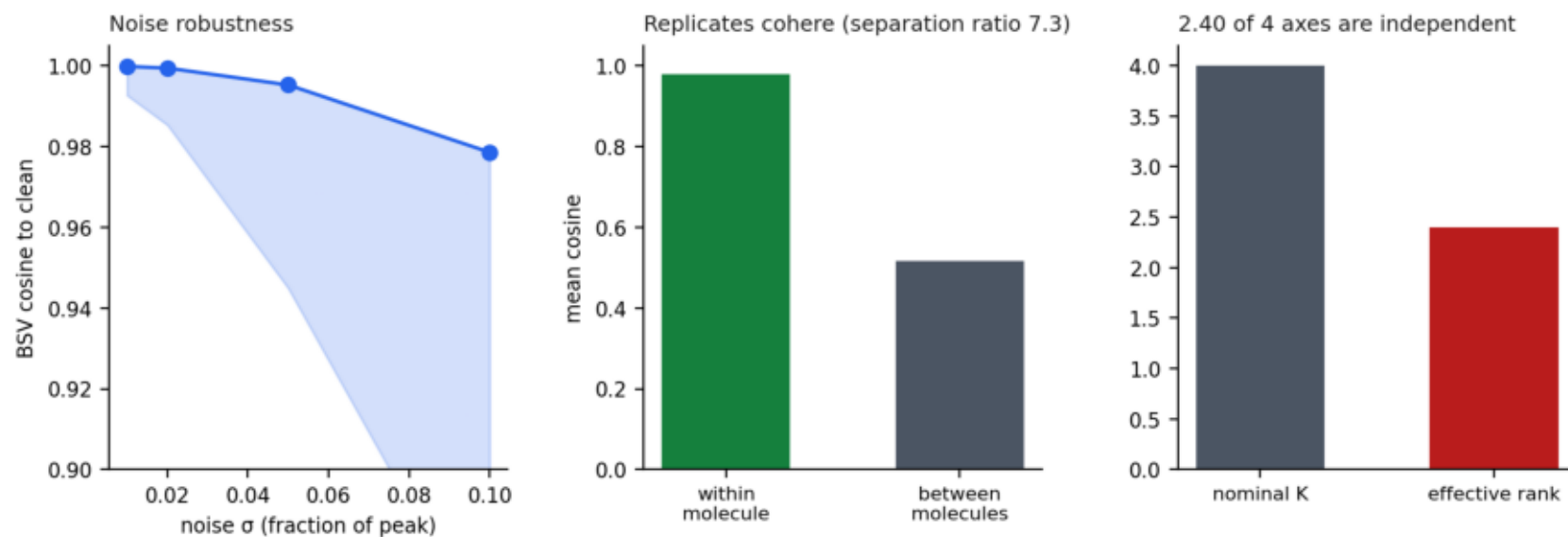
Out-of-domain detection. A non-negative dictionary of Raman motifs reconstructs Ag-SERS of the same metabolites comfortably, so the atlas cannot tell modality. Reported as a failed gate, not compensated.



THE FAILING GATE. On real Ag-SERS the OOD score is at chance, and SERS spectra are BETTER explained than the references. A non-negative dictionary of Raman motifs reconstructs SERS of the same metabolites comfortably, so the atlas cannot tell modality. The synthetic probe reaching 0.946 is precisely why the real probe was necessary.

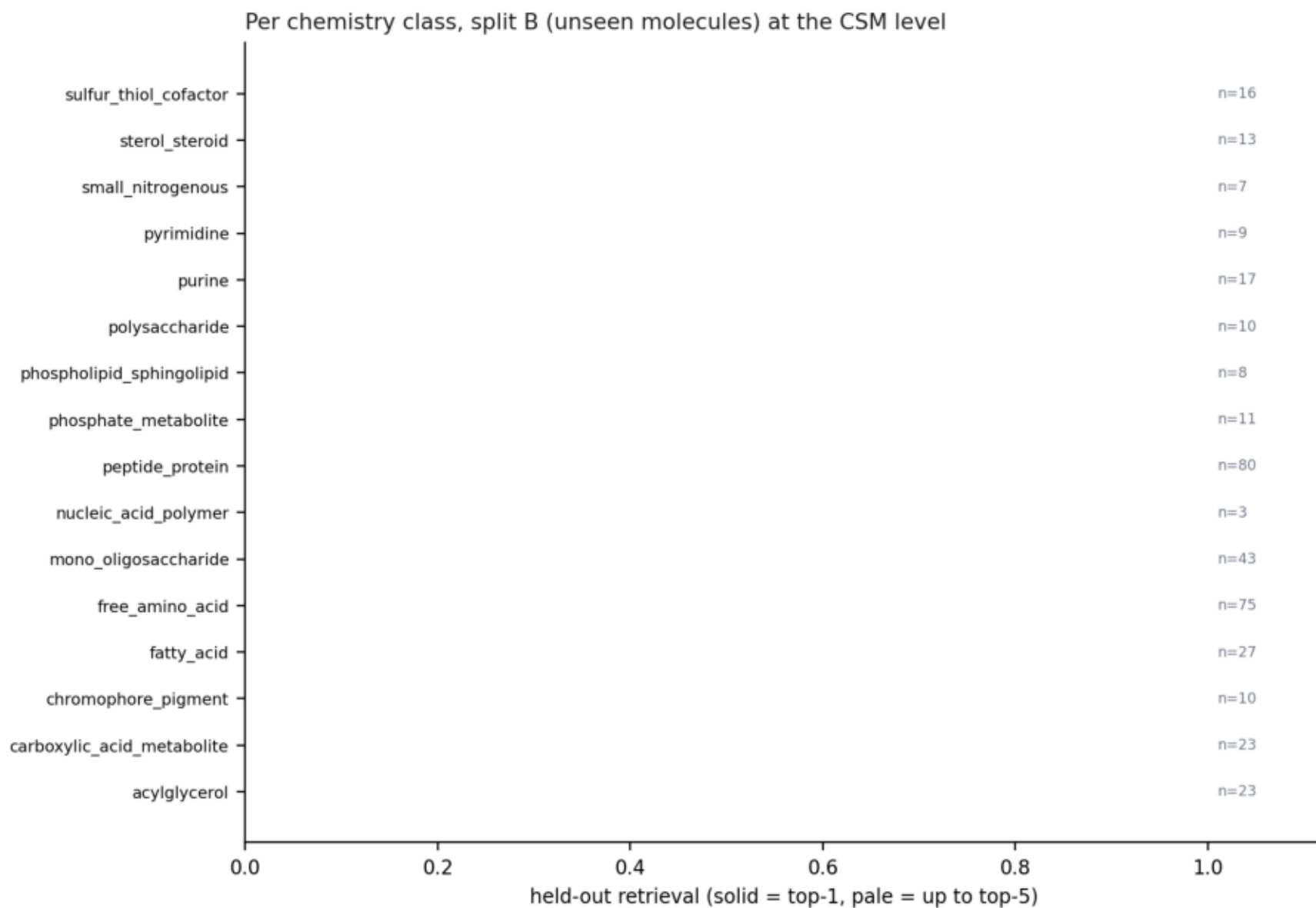
Figure 09

The Biochemical State Vector — absolute, non-negative, four axes. Effective rank is reported alongside K (risk R-12).



The Biochemical State Vector: absolute, non-negative, four axes, effective rank 2.40. Replicates cohere at cosine 0.979 with a between/within separation ratio of 7.26, and the vector survives noise at $\sigma = 0.05$.

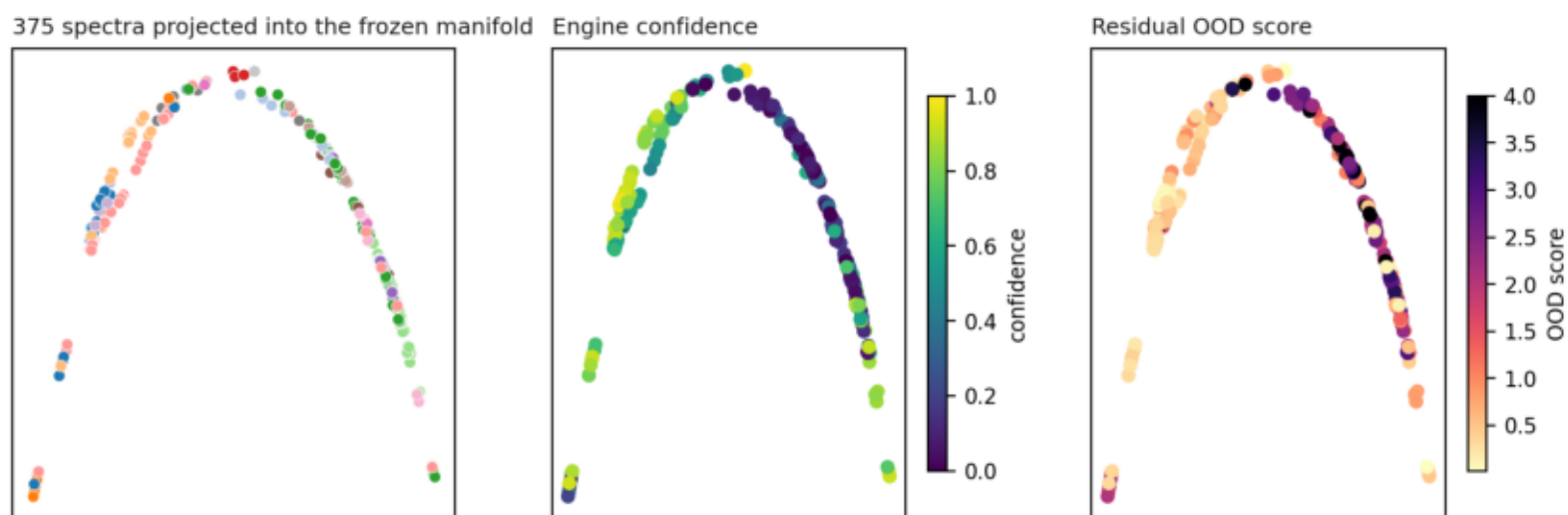
Figure 10



Per chemistry class, unseen molecules, at the CSM level.

Figure 11

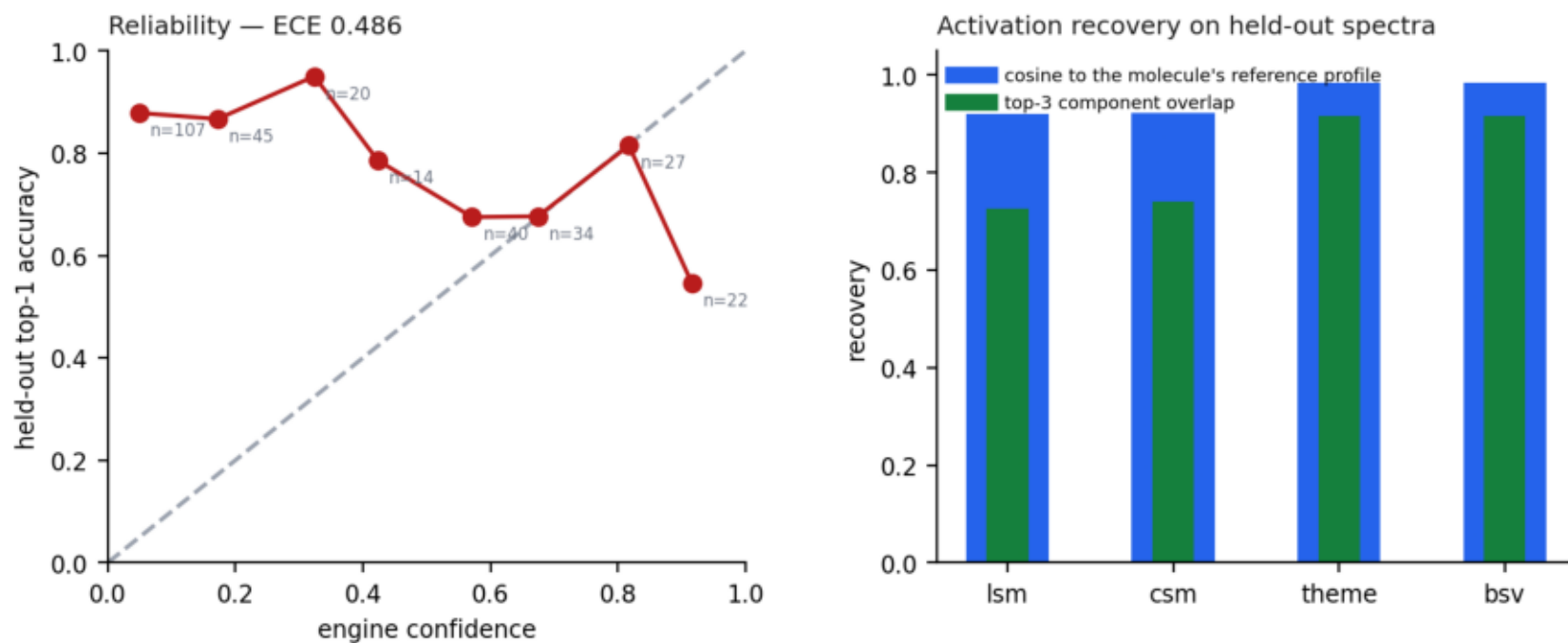
Geometry projection. Neighbourhood purity 0.482 at 4.1× chance.



All 375 reference spectra projected into the frozen manifold, coloured by chemistry, by engine confidence and by OOD score. Neighbourhood purity is 4.06x chance.

Figure 12

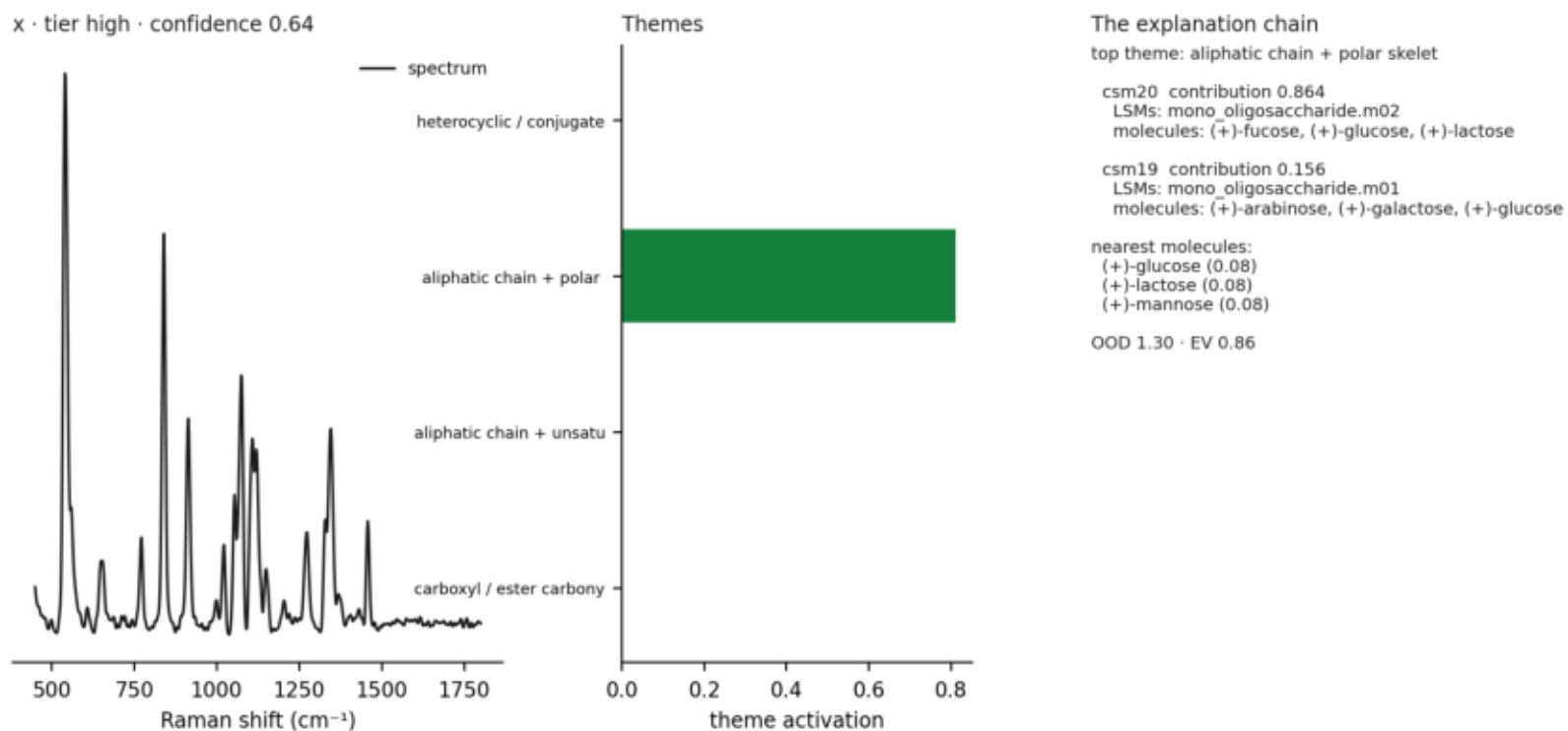
Confidence is poorly calibrated — it is monotone but overconfident, and that is a reported failure.



Confidence is monotone in accuracy but badly overconfident (ECE 0.486) — a reported failure. Activation recovery against each molecule's own reference profile rises with abstraction.

Figure 13

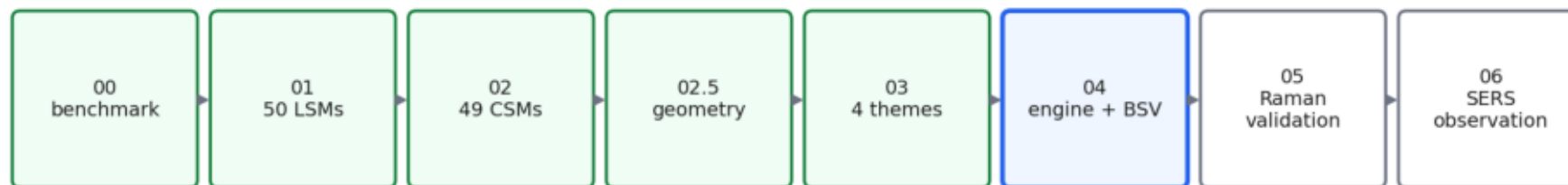
A worked example. Every activation resolves to CSMs, LSMs and named molecules — nothing opaque.



A worked example end to end, with the explanation chain resolved from the frozen registries at call time so it can never disagree with the atlas.

Figure 14

GAIRA V7 architecture after Phase 04



Engine frozen. BSV dimension 4 (effective rank 2.40). Held-out: molecule top-1 0.799, class top-1 0.855 at the CSM level.

Open: OOD cannot separate real Ag-SERS (AUROC 0.55); confidence poorly calibrated (ECE 0.49); dictionary leakage +0.055 top-1.